Week 3 – Exploring & Understanding Inversion

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Duration: 3 hours **Setting:** Lecture room with laptops (sphere inversion app)

Purpose

This workshop introduces the structure of inverse problems in potential-field geophysics, contrasting forward prediction with inverse estimation. Students explore the role of misfit, data weighting, and regularization, and examine how non-linearity and non-uniqueness arise in gravity modelling. Through a sequence of activities, they classify linear vs. non-linear models, reason qualitatively about non-uniqueness, and engage in hands-on inversion of a buried sphere using grid search and Gauss–Newton methods.

Learning Outcomes

- Distinguish forward and inverse problems and articulate the role of misfit, weighting, and regularization.
- Classify simple geophysical relations as linear or non-linear and propose appropriate linearizations.
- Recognize manifestations of non-uniqueness in gravity and magnetic profiles and propose discriminating data.
- Derive or confirm analytical Jacobian expressions for the buried sphere model.
- Interpret misfit surfaces, inversion paths, and the effect of noise and damping.
- Explain parameter trade-offs and resolution limitations inherent to the buried-sphere problem.

Reference concepts align with Week 3 notes on forward/inverse modelling, misfit structure, and non-linear least squares.

Roles and Grouping

Create mixed groups (4–5 students). Recommended roles: *Facilitator, Analyst, Recorder, Skeptic, Reporter*. Rotate roles between major activities.

Session Flow (180 min)

0–30 min: Mini-lecture — Forward vs Inverse, Misfit, Weighting, Challenges

- Forward vs. inverse models: $m \mapsto g(m)$ vs. $d \mapsto m$.
- Misfit: residual $r = d - g(m)$; weighted least squares $\Phi_d = \|W_d r\|_2^2$.
- Role of W_d (noise) and W_m (model constraints); damping λ.
- Challenges: non-linearity (Jacobian needed), non-uniqueness (multiple m give similar $g(m)$), ill-conditioning.

30–60 min: Activity — Linear or Non-linear?

Using the ten-equation handout:

- Identify model parameters and classify each relation as *linear*, *non-linear*, or *linearizable*.
- If non-linear, propose a linearization (log-transform, re-parameterization, Taylor expansion/Jacobian).
- Deliverable: brief notes per group.

70–90 min: Activity D — Non-uniqueness (Jigsaw)

Curves-only reasoning:

- Scenario A: gravity profiles — two distinct geometric models can reproduce similar anomalies.
- Scenario B: magnetic dike profiles — width/depth/magnetization trade-offs.
- For *each* curve: propose at least two plausible geologic/geometric models.
- List discriminating data (station density, gradients, susceptibility/remanence, multi-component data, mapping).
- Deliverable: 1-paragraph argument explaining which model to pursue first and what data to collect next.

90–100 min: Buried Sphere — Introduction

- App overview: Data & Model view; Misfit view; controls for acquisition, noise, synthetic model.
- Construct a true model & generate data (freeze a single noise realization).

100–120 min: Buried Sphere — Jacobian Derivatives

1. Students derive/confirm analytical expressions for $\partial g/\partial z$, $\partial g/\partial R$, $\partial g/\partial \Delta\rho$.
2. Compare with the app's Jacobian; discuss relative sensitivities: amplitude $\propto R^3\Delta\rho$, width controlled by depth z.

120–155 min: Buried Sphere — Investigation

- **Grid search:** Explore pairwise misfit slices and 3-D misfit surface. Identify local vs global minima.
- **Inversion path:** Run Gauss–Newton with different λ; examine step length and stability.
- **Effect of noise:** Increase σ; observe widening misfit basins and parameter uncertainty.
- Discuss links to curvature, conditioning, and resolution.

155–170 min: Buried Sphere — Non-uniqueness

- Demonstrate parameter trade-offs (e.g. $R^3\Delta\rho$) and resolution ellipses.
- Compare inversion results with misfit contours; highlight underdetermined directions.

170–180 min: Wrap-up

- Key takeaways: forward/inverse distinctions, misfit and weighting, non-linearity, non-uniqueness, damping.
- Open Q&A; preview of next week's material.

Materials

- Linear/non-linear handout or slide deck.
- Activity D handouts (gravity and magnetic curves-only plots).
- Buried sphere app (laptop per group).
- Whiteboard or shared notepad for Jacobian derivations.

Assessment

- Activity notes (classification & argumentation).

- Jacobian derivation steps.
- Short reflection or exit-ticket on non-linearity and non-uniqueness.